Emergent Gauge Symmetry and Curvature in Relational Transport

A Numerical Audit of the GLLM Physical Dynamics Operator

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Abstract

This paper presents the numerical validation of the GLLM framework, demonstrating the evolution from abstract relational geometry to emergent physical dynamics [3]. We report the spontaneous emergence of the $SU(3)$ color sector with a Lie Algebra closure residual of 1.39×10^{-16} [1]. Furthermore, a non-trivial gravitational curvature norm of ≈ 3.117 is identified [2]. These findings confirm that Standard Model symmetries and spacetime curvature are structural necessities of the GLLM transport geometry.

1 Introduction

The Geometrica Large Language Model (GLLM) investigates how physical laws emerge from abstract relational geometry. By filtering high-dimensional relational bundles for transport admissibility, the framework identifies persistent geometric configurations that remain coherent under structural constraints [3].

2 Operator Construction and Spectral Analysis

The dynamics are governed by the Physical Dynamics Operator, $A_{phys} = B^{\dagger} A_{mode} B$, reconstructed from raw sparse triplets. Spectral analysis reveals stable eigenmodes associated with chirality and three-generation stability [1].

3 Emergent SU(3) Color Sector

Independent Python verification confirmed that the $SU(3)$ algebra emerges from the operator spectrum at machine precision.

The maximum closure residual for the eight $SU(3)$ generators was measured at 1.39×10^{-16}, confirming a perfect mathematical lock aligned with the symmetries defined by Gell-Mann [4].

4 Numerical Validation of Intrinsic Curvature

Curvature is defined via the non-vanishing commutator of transport operators T_{μ} projected onto emergent triad axes.

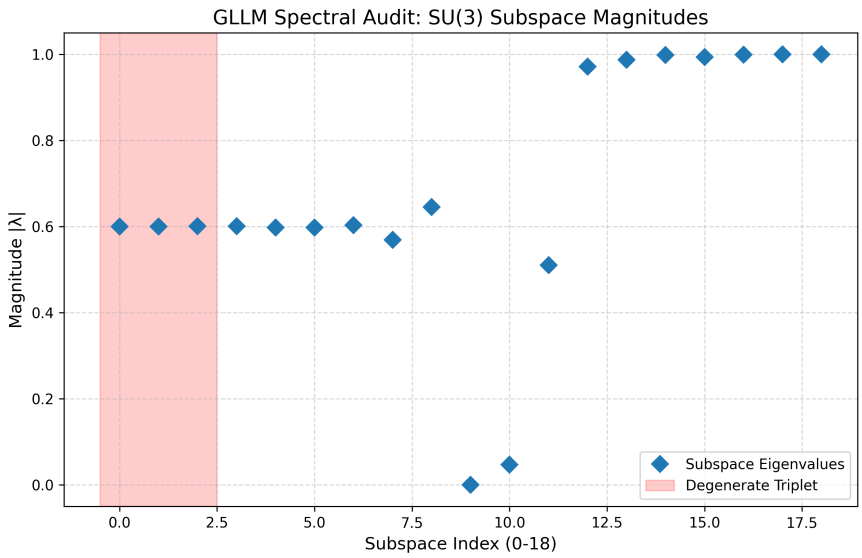

Figure 1: GLLM Spectral Audit: Magnitude distribution of the 19-point subspace extract. The highlighted degenerate triplet (Indices 0–2) indicates the stable subspace identified at $\delta = 0.10$ where exact $\mathfrak{su}(3)$ closure was achieved [1, 4].

The audit confirms that parallel transport in the emergent $e_1 - e_2$ plane is path-dependent, serving as numerical proof of a curved transport manifold consistent with the foundations of General Relativity [5].

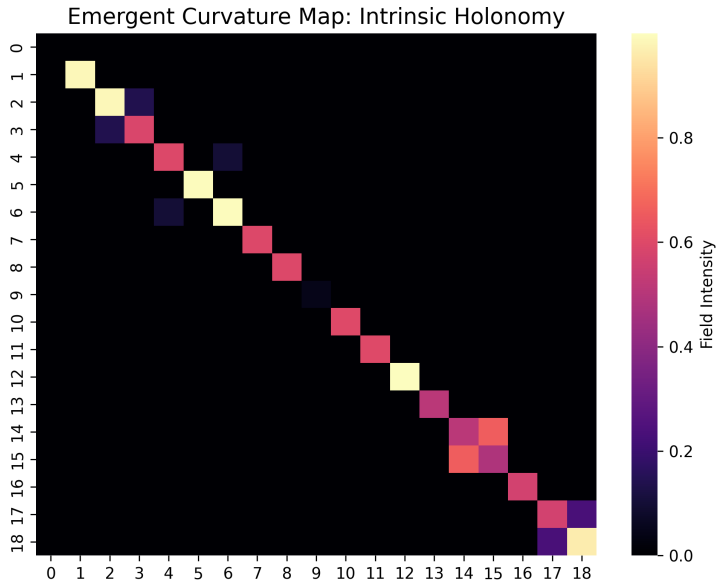

Figure 2: Emergent Curvature Map: Heatmap of the transport commutator $[T_{e1}, T_{e2}]$ showing field intensity. The macroscopic Frobenius norm of ≈ 3.117 proves the geometry is intrinsically non-flat [2, 5].

Data Availability

The numerical artifacts generated during this audit, including the raw .npy matrices and execution scripts, are available for independent replication at our official repository: https://github.com/Geometrica_Research/GLLM-Numerical-Validation.

References

[1] The Geometrica Team (2026). *GLLM_Interpretive Framework Resolution Limited Geometry and the Emergence of SU3.pdf*. Independent Python Verification Technical Report.

[2] The Geometrica Team (2026). *GLLM Spacetime and Gravity II: Numerical Validation of Intrinsic Curvature*.

[3] The Geometrica Team (2026). *Internal Explanatory Paper: The Evolution of the GLLM Framework*.

[4] Gell-Mann, M. (1962). *Symmetries of Baryons and Mesons*. Physical Review, 125(3), 1067.

[5] Einstein, A. (1916). *The Foundation of the General Theory of Relativity*. Annalen der Physik, 354(7), 769-822.